

SensARy Substitution: Augmented Reality Techniques to Enhance Force Perception in Touchless Robot Control

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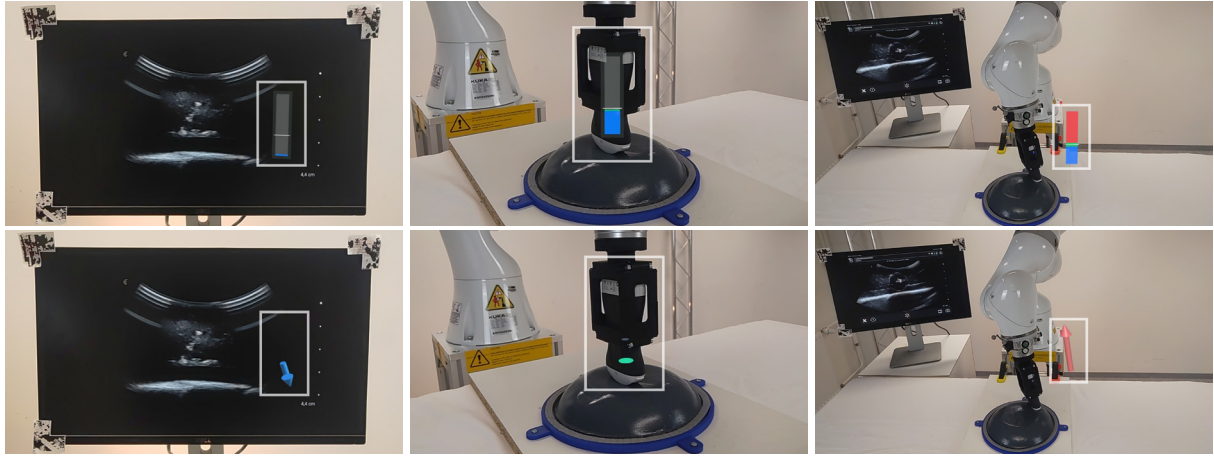


Fig. 1: Overview of force visualization types and placements. The top row shows bar visualizations, and the bottom row shows arrow visualizations, both at different force levels (left to right: too low, appropriate, and too high). The left column shows placement on the screen displaying the sonographic image, the middle column shows placement on the ultrasound probe in contact with the phantom, and the right column shows placement anchored to the user's field of view.

Abstract—The lack of haptic feedback in touchless human-robot interaction is critical in applications such as robotic ultrasound, where force perception is crucial to ensure image quality. Augmented reality (AR) is a promising tool to address this limitation by providing sensory substitution through visual or vibrotactile feedback. The implementation of visual force feedback requires consideration not only of feedback design but also of positioning. Therefore, we implemented two different visualization types at three different positions and investigated the effects of vibrotactile feedback on these approaches. Furthermore, we examined the effects of multimodal feedback compared to visual or vibrotactile output alone. Our results indicate that sensory substitution eases the interaction in contrast to a feedback-less baseline condition, with the presence of visual support reducing average force errors and being subjectively preferred by the participants. However, the more feedback was provided, the longer users needed to complete their tasks. Regarding visualization design, a 2D bar visualization reduced force errors compared to a 3D arrow concept. Additionally, the visualizations being displayed directly on the ultrasound screen were subjectively preferred. With findings regarding feedback modality and visualization design our work represents an important step toward sensory substitution for touchless human-robot interaction.

Index Terms—Augmented Reality, Sensory Substitution, Force Perception, Human-Robot Interaction, Robotic Ultrasound.

1 INTRODUCTION

In human-robot collaboration, the precision and repeatability of robotic actuators are combined with the cognitive abilities and flexibility of humans to enable adaptability to dynamic work environments [1]. Although manual control of different robot functions allows for the operator's expertise to be integrated into the robot control, the loss of direct perception poses a challenge [36]. Therefore, the integration of haptic feedback has proven to be a valuable tool to improve situational awareness by providing tactile cues [11]. While haptic input devices can provide direct feedback by mimicking the robot's sensory responses, these modalities are not always feasible due to task-specific constraints such as spatial limitations or associated high costs [32, 36]. Another method to provide haptic perception is through sensory substitution,

where information is transcoded from one sensory modality to another, such as auditory [21], vibrotactile [6] visual [34] or multimodal [19] stimulation. In contrast to direct force feedback, sensory substitution offers the advantage of small forces being perceived at high stimulation levels without impacting the operator's motions and, therefore, without compromising system stability [31]. Sensory substitution in robotics has been shown to reduce the overall force applied [15], help maintain consistent force [46], and improve stability and precision [53] even when direct haptic feedback is provided.

In visual sensory substitution, the proper positioning of force visualization is crucial to allow for simultaneous focus on the task and accurate application of force [9]. This can be facilitated by superimposing the visual sensory substitution on the direct view of the physical interaction space using Augmented Reality (AR). In the field of human-robot interaction, implementing AR interfaces is of increasing research interest and has shown promising results [29, 45]. This potential, thus, merits further investigations on the design and positioning of force visualizations in AR.

In general, the potential for sensory substitution to enhance force perception was explored for tasks like teleoperation [37] or robotic surgery [37]. However, less research has been conducted on the enhancement of force perception during robotic ultrasound, where the

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focus lies not only on the robotic actuator but also on the sonographic data, limiting the transferability of prior work. Robotic ultrasound uses a robotic arm to position the probe, aiming to reduce the workload on sonographers by partly automating the imaging process [27]. During ultrasound imaging, acoustic coupling requires a force to be applied to the tissue, which in turn significantly influences tissue deformation. Variations in applied force can consequently affect imaging repeatability and quality, posing challenges in reliable image-based measurements [5, 13, 50]. Applying a constant force has been shown to increase reliability and mitigate measurement errors [22]. While robotic ultrasound can be intuitively enabled through touchless interaction, maintaining adequate force during operation remains a challenge [43].

Therefore, in this work, we compare different AR-based sensory substitution techniques derived from existing approaches in other domains to assess their suitability for facilitating force perception in robotic ultrasound. By building on established methods, we aim to incorporate previous findings into our work and focus on the systematic comparison of multiple potentially influencing factors. Since related work highlights the distinct advantages of visual [16, 24], vibrotactile [6], and multimodal [19] feedback modalities, this study aims to explore the potential of both unimodal visual and vibrotactile and multimodal feedback. Given the importance of both the design and the positioning of visual sensory substitution, we have implemented two visualization approaches in AR displaying them at three distinct positions. Through evaluating feedback modality, visualization type, and position, we contribute specific insights into the effectiveness of different feedback approaches for conveying information about contact force in human-robot interaction. To the best of our knowledge, no prior comparative evaluation of AR placements or the combination of these factors has been conducted. This allows us to understand how these factors interact and impact user performance in a practical application, providing insights that cannot be inferred from studies focusing on isolated factors. From these insights, guidelines for sensory substitution in human-robot interaction are derived. To enable generalizability to AR interfaces beyond robotic ultrasound, the applicability of these guidelines and potential adaptations to enhance their transferability to other tasks with limited direct force perception, such as medical or industrial robotic applications, are discussed.

2 RELATED WORK

This section outlines feedback modalities, visualization concepts, and placements in the broader field of sensory substitution for conveying force information, including applications beyond robotic ultrasound. In robotic ultrasound, the axial force along the longitudinal axis of the probe is critical for achieving acoustic coupling and ensuring image quality [20, 27]. It is also the primary direction of force during ultrasound procedures [12]. Therefore, this study focuses on techniques that effectively convey the force in one direction, excluding multi-dimensional force representations as they introduce additional complexity that is not required for this task. Given that this study also aims to investigate both the modalities and spatial placement of force feedback, visualizations that are inherently anchored to a fixed position are also excluded to allow greater flexibility in exploring different visualization placements.

2.1 Sensory Substitution Modalities

Several research groups have conducted comparative studies on multimodal and unimodal approaches to assess the suitability of different modalities for sensory substitution conveying force information. Jonetko et al. [21] compared three methods of force visualization with auditory feedback for grasping tasks. While all feedback modalities reduced grasping force, no performance differences were found across conditions. Moreover, although users preferred multimodal, redundant feedback, it appeared to be less efficient. Since continuous auditory feedback might be disturbing in an already noisy intervention room and potentially distracting communication between medical staff discrete auditory feedback may be more promising. Kitagawa et al. [24] compared visual force feedback with discrete auditory feedback for a robotic suturing task. Their results indicated visual outperforming

auditory feedback in terms of force consistency. Haruna et al. [16] investigated visual, auditory, and vibrotactile feedback to signal excessive force during a grasping task. Their results revealed that visual feedback was the most effective in reducing grasping force without increasing cognitive load. Chan et al. [6] explored force augmentation for robot trajectory planning using visual and vibrotactile feedback. Their findings suggested providing unimodal vibrotactile sensory substitution resulting in smaller force errors than visual or multimodal feedback. In contrast, Howard and Szweczyk [19] observed that visual feedback had an overall better performance when compared to different vibrotactile feedback types for force representation in laparoscopy. Additionally, they found that multimodal feedback enhanced task precision compared to visual and tactile feedback alone.

Although these studies comparatively evaluate different sensory substitution techniques and their results suggest that visual and vibrotactile techniques are suitable for force display, their findings are not specific to robotic ultrasound. Furthermore, these studies do not consider the effects of different force visualization designs and positioning on the need for multimodal redundant vibrotactile and visual sensory substitution.

2.2 Visualization Concepts

Different designs of visual sensory substitution to facilitate force perception have been explored not only for human-robot interaction but also in areas such as minimally invasive surgery or remote collaboration. The visualization concepts include: colored areas [2–4, 21, 23, 40], bars indicating the force only by height [19, 23, 42, 52] or by height and color change [3, 4, 15, 24, 34, 46, 53], Arrows [6, 9, 18, 21], Numbers [9, 21], Heat-Maps [3], by the size of a highlighted area [14] or the display of a force gauge [13]. While most of these visualizations only convey information on force magnitude, Horeman et al. [18] as well as Zhu et al. [53] additionally integrated visualization of force direction. Black et al. [4] conducted a comparison of two of these force visualizations in a remote collaboration task for ultrasound. The visualizations included continuous color changes of a virtual probe and an error bar adjusting size and color based on the current force offset. The study results indicated the superiority of the error bar, with all users preferring this visualization, as the disappearing bar was easier to perceive than detecting continuous color changes. Similarly, Kim et al. [23] compared two approaches for sharing hand force in remote collaboration, finding that a size-changing bar provided clearer force visualization than a superimposed colored mesh on the remote collaborators' hands. Aviles-Rivero et al. [3] compared four force visualizations for robotic surgery suggesting several design implications: overlapping information over the region of interest and overburdening the display with unnecessary text should be avoided; small visualization should be placed as close as possible to the region of interest; color coding and simple yet efficient geometry should be used and multiple cues that represent information, such as combining position and color should be provided.

While these studies suggest that redundant cues such as color and varying size enhance force perception, they do not directly compare the two prevalent visualizations, namely an arrow and a bar. Furthermore, only visualizations depicting the magnitude of the force are compared, without considering visualization designs that provide additional information about the force direction.

2.3 Visualization Placement

The placement of AR content to enhance situational awareness has been shown to impact reaction time and comprehension [51]. Given that most existing literature on force visualizations focuses on robotic surgery [37], the region of interest is typically visible through the endoscopic image displayed on a screen. This screen is then utilized for presenting the force feedback, either adjacent to the image [3, 18, 24, 42, 46] or directly overlaid onto the image at the position of the instruments [2, 3, 14, 15, 19, 40]. Similarly, Yusof et al. [52] and Moortgat-Pick et al. [34] investigated teleoperation for grasping tasks, displaying the remote area on a monitor and integrating force feedback alongside the video stream. Gilbertson and Anthony [13] studied the impact of force visualization in freehand ultrasound examinations, displaying a force gauge beside the ultrasound image shown on

an external monitor. In contrast to these screen-stabilized approaches Jonetzko et al. [21] as well as Chan and Quintero et al. [6] investigated robot-stabilized placement of visual sensory substitution. They utilized the Microsoft HoloLens to display force visualizations directly at the position of the robot end-effector. De Franco et al. [9] compared screen and robot-stabilized visual force feedback for a polishing task. While the former displayed real-time force plots on a desktop monitor, the latter utilized a holographic arrow displayed via the HoloLens. Results indicated that the robot-stabilized interface promotes more consistent force maintenance, while participants encountered difficulties focusing on the task with desktop feedback. While both screen- and robot-stabilized visualization placements anchor the information at world positions either in the direct or the remote environment, Plabst et al. [38] expanded their study beyond world-stabilized positions, investigating the impact of head, body, and world-stabilized information on notification visualization. They found wrist notifications promising for high interaction scenarios, while world-stabilized notifications are suited for stationary tasks. Fixed notifications in the user's field of view (FOV) are preferred for tasks involving attention shifts. Rzayev et al. [41] found that while head-stabilized information reduced response times, it increased distraction compared to world- and body-stabilized displays. Similarly, Li et al. [28] showed that object stabilization improved user experience by minimizing visual interference. Chua et al. [7] identified the center-right FOV position as optimal for balancing performance and usability in head-stabilized interfaces. Lee and Woo [25,26] also found that displaying notifications in the middle-right FOV reduced response time and task load.

These studies compare screen-stabilized with robot-stabilized force displays and world-stabilized with body-stabilized notification and user interface displays, yet they do not comprehensively compare all three possible display positions. This gap is particularly relevant for applications such as robotic ultrasound imaging, where the region of interest consists of both, the robot positioning the probe and the screen visualizing the sonographic image.

3 TECHNICAL METHODS

In ultrasound imaging, it is crucial to both avoid excessive force for patient safety and maintain a constant force for consistent tissue deformation. Therefore, the implemented sensory substitution methods should not convey the absolute force but rather the applied force in the axial direction of the probe, expressed as the relative force to a target force (F_{target}). The previous section presented several sensory substitution concepts for force perception. However, these findings do not investigate the interaction between different visualization placements, designs, and multimodality, and are additionally not specific to the context of robotic ultrasound. To bridge this gap, we implemented two visual concepts, one 2D and one 3D, each representing relative force deviation through shape and color changes, along with a vibrotactile concept displaying force through different patterns and frequencies (see Figure 2). For exploring different visualization placements, a Head-Mounted Display (HMD) was employed. The HMD enables flexible prototyping of positions with consistent visual properties, ensuring a controlled environment for evaluating the impact of visualization placement on user performance. The resulting visual sensory substitution techniques are presented in the supplementary video.

3.1 Visualization Concepts

3.1.1 Bar

Given that using bars with heights proportional to the applied force is a prevalent method in the literature for visual force feedback, the first visualization features stacked bars (see top visualization in Figure 2 and top row of Figure 1). The currently applied force is continuously indicated by the height of the 2D bar, while discrete color coding indicates its relation to F_{target} . The color scheme is defined as follows:

$$color_{Bar} = \begin{cases} \text{blue} & \text{for } F < F_{target} - 1\text{N} \\ \text{green} & \text{for } F_{target} - 1\text{N} < F < F_{target} + 1\text{N} \\ \text{red} & \text{for } F > F_{target} + 1\text{N} \end{cases}$$

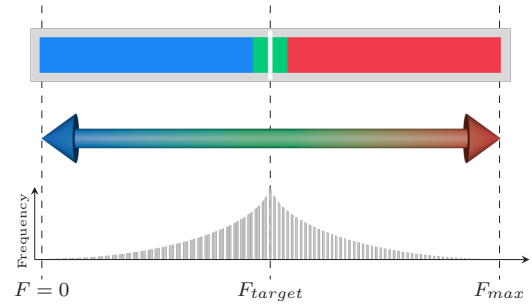


Fig. 2: Different sensory substitution concepts.

Additionally, a white line marks the goal force F_{target} . The threshold of 1 N above or below F_{target} was empirically tuned aiming to provide users with precise feedback to distinguish small deviations in applied force. The visualization's orientation is continuously adjusted to be facing towards the observer.

3.1.2 Arrow

In contrast to the first visualization, where size represents the force magnitude, another approach found in the literature maps the size to force deviation from F_{target} , causing the visualization to disappear when the target force is reached. To additionally convey not only force magnitude but also direction, the second visualization employs a 3D arrow varying in height, color, and pointing direction (see bottom row in Figure 1). The orientation of the arrow is adjusted based on the direction of the applied force: for forces smaller than F_{target} , the arrow points in the direction of the applied force, with its color interpolated between blue and green. Conversely, for forces larger than F_{target} , the arrow points in the opposite direction of the applied force, with its color interpolated between green and red. Therefore, the pointing direction always indicates the direction in which movement is needed to attain F_{target} . The scaling of both the length and the color interpolation is linear (see middle visualization in Figure 2).

3.2 Visualization Placement

Three distinct positions of the visualization are explored (see Figure 3). The first position is a screen-stabilized placement commonly used in the literature, anchoring the visualization to the screen displaying the ultrasound image (*screen* position). The screen's position is determined using image markers attached to its corners, which are detected once and pinned at its position. Considering the relevance of both the screen and probe positions during the interaction, and based on promising results regarding robot-stabilized visualization placement, the second placement is *probe* position. Here, virtual content is superimposed onto the ultrasound probe, with the probe's position continuously derived by the robot using forward kinematics and transformed through an initial registration at the beginning of the interaction. As the intervention focuses on the probe's general position on the body and its contact point, placing the visualization at the upper part of the probe is assumed not to obstruct the view. The third position aims to explore body-stabilized information. Considering limited interaction with virtual content during the task and potential shifts in attention between the ultrasound probe and the screen, following Plabst et al. [38], the third visualization is anchored to the user's head (*FOV* position). Based on the findings of Lee and Woo [25,26] as well as Chua et al. [7], the virtual content is positioned in the center-right for both the *Screen* and *FOV* positions. To ensure good visibility without obstructing relevant information, the virtual content is displayed directly adjacent to the ultrasound image for screen placement. For the *FOV* position, the virtual content is placed 1 m from the user, vertically centered, and positioned within the right quarter of their field of view.

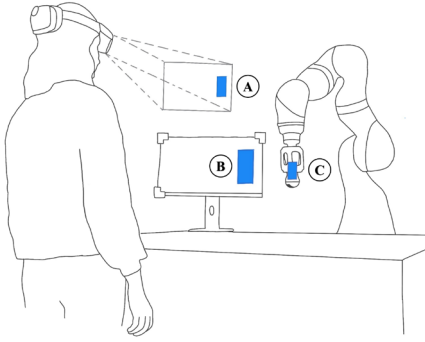


Fig. 3: Overview over different force visualization placements: (A) FOV, (B) Screen, and (C) Probe.

3.3 Vibrotactile Feedback Concept

Based on the findings reported by Howard and Szwedczyk [19], which suggest that pulsed vibrotactile feedback performs better than continuous feedback with varying amplitudes, the vibrotactile feedback in this study utilizes vibration pulses with varying pause lengths to convey deviation from F_{target} . Following the work of Chan et al. [6], different vibration patterns are used to indicate forces that are higher or lower than F_{target} . Specifically, forces smaller than F_{target} are represented as a double pulse with a pause of 10ms between the pulses, while forces higher than F_{target} are represented as a single pulse. To enhance the detectability of differences in forces closer to F_{target} and facilitate faster detection, a scaling approach similar to that used by Marquard et al. [30], as described by Equation 3, is employed. This approach was selected because it adapts a widely known logic used in car parking systems: as the target force is approached, the pause length T_{pause} decreases, creating a faster vibration pattern that becomes most rapid at F_{target} . Using the highest vibration pulse frequency near F_{target} , rather than increasing the frequency proportional to the force (as suggested by Howard and Szwedczyk [19]), was chosen to provide the highest information density close to F_{target} , making deviations easily detectable. While this contrasts with the linear scaling used for the visual force representations, the information density is assumed to be comparable across modalities, since the visualizations present multiple channels of information through shape and color changes, with, for example, the bar and its green area making small changes around F_{target} easily perceptible. T_{pause} is scaled to be between the minimal pause length $T_{\text{min}} = 10\text{ms}$ at F_{target} and the maximal pause length $T_{\text{max}} = 100\text{ms}$ at the maximal allowable force F_{max} using a scaling factor K derived from the relative force F_{rel} .

$$F_{\text{rel}} = \min \left\{ 1, \frac{F_{\text{curr}} - F_{\text{target}}}{F_{\text{max}} - F_{\text{target}}} \right\} \quad (1)$$

$$K = \sqrt{-(F_{\text{rel}} - 2) \cdot F_{\text{rel}}} \quad (2)$$

$$T_{\text{pause}} = K \cdot (T_{\text{max}} - T_{\text{min}}) + T_{\text{min}} \quad (3)$$

The vibrotactile feedback is delivered through a vibrotactile wristband worn on the dominant hand, with the patterns transmitted via an auditory signal. Therefore, Equation 3 is utilized to scale T_{pause} in between simple inaudible sounds with a duration of 100 ms at a frequency of 100 Hz created using the Windows Beep function.

3.4 Force Data

The force data is derived from the robotic manipulator by utilizing its joint torques. The force thresholds are set in alignment with the values reported by Dhyani et al. [10] for patients with a high BMI, with $F_{\text{target}} = 10\text{N}$ and $F_{\text{max}} = 30\text{N}$. Additionally, F_{max} serves as a safety measure threshold, triggering an emergency response if exceeded. In such cases, the interaction is halted, and the robot is moved 10 mm in the opposite direction of the force. Due to the relatively small forces and

the low load consisting of the ultrasound probe, there is susceptibility to digital noise affecting the cartesian force estimation. To mitigate this noise, a Kalman filter [48] is employed. This filter adjusts new measurements based on previous ones, with the filter weights being empirically tuned.

3.5 Robot Control

The robot is controlled using a touchless interaction technique that integrates hand gestures to manipulate the position and orientation of the robot end effectors. Employing the metaphor of a virtual dummy probe, relative hand movements are transferred to the dummy probe and subsequently mapped to corresponding robot movements. During the interaction, rotation, and translation can be manipulated either simultaneously or separately, with the corresponding modes selected via a foot pedal. The interaction initiates upon pressing another pedal, at which point the virtual dummy probe aligns with the orientation of the real probe in the user's hand. To facilitate this interaction, as well as the placement of the force visualization at the probe and the orientation of the arrow visualization, a registration between the HMD and the robot is required. This registration is achieved using image tracking to pivot calibrate a marker attached to the robot end effector, followed by paired-point registration, similar to the mid-air calibration method described by Mielke et al. [33].

3.6 System Setup

The system setup comprises a KUKA LBR IIWA (KUKA AG, Germany) robotic arm with a Clarius (Clarius Mobile Health, Canada) ultrasound probe attached via a custom 3D-printed mount (see Figure 1). Due to its frequent use in human-robot interaction [8], a Microsoft HoloLens 2 (Microsoft Corporation, USA) is used as HMD, displaying virtual content and facilitating interaction. Interaction is further employed by a foot pedal with two switches. Vibrotactile feedback is provided by a Lofelt Basslet (Lofelt, Germany) bracelet, a haptic wristband with a high-bandwidth vibration actuator, driven by auditory signals transmitted via Bluetooth. The system consists of two Unity 3D applications. The first application runs on the HoloLens, integrates registration with the robot, and determines the position of the ultrasound screen using the Vuforia SDK (PTC Inc., USA), visualizes the force displays, and maps hand gestures to robot movements. The second application runs on an external PC, managing communication between the HoloLens and the robot, filtering force data, and for the vibrotactile condition, translating force into inaudible auditory signals sent to the bracelet. For temporal alignment between visual and vibrotactile feedback, both feedback modalities are updated at the same frame rate. Additionally, a program implemented in KUKA Sunrise.OS runs on the robot controller, utilizing the KUKA Sunrise.Servoing package for real-time movement control. Communication between the three aforementioned components occurs via UDP connection.

4 EVALUATION METHODS

4.1 Task

The task of the study was to integrate ultrasound imaging to guide the probe to specific positions while maintaining reproducible tissue deformation by keeping the force constant at a predefined level. Thus, a homemade agar-agar phantom was constructed, containing three distinct geometric shapes (a sphere, a pyramid, and a cube) made of rubber. To require both translational and rotational movements, the phantom was constructed in the shape of a semi-sphere, with the lateral gradient requiring the probe to be rotated to maintain image quality. Figure 1 depicts the appearance of the phantom, alongside corresponding sonographic images of the geometric structures. The participants were tasked with visualizing the three target structures successively while maintaining a constant force as close as possible to F_{target} . To achieve this, the robot was initially moved from its mid-air starting position to the surface. Participants then maneuvered the probe along the surface to identify an acoustic window where the outline of the geometry was clearly visible before moving on to the next target structure. To prevent small initial forces before the first surface contact from influencing the study results, participants were instructed

to move the probe towards the surface and apply pressure until F_{target} was attained at the onset of each task. Upon reaching F_{target} , a blue LED on the robot would blink, signaling the start of force recording.

4.2 Participants

Although clinical expertise was not necessary due to the abstract nature of the task, a basic understanding of sonographic images was required for task completion. Therefore, medical students from the local university were invited. They received 30€ for their participation. In total, Twenty-one (21) medical students participated in the study (15 female, 6 male) with ages ranging from 22 to 32 years (mean (M) = 26.14, standard deviation (SD) = 3.23). They were in their 3rd to 5th year of university (M = 4.41, SD = 0.70). None of the participants reported any form of color vision deficiency. Participants reported their familiarity with ultrasound (M = 3.48, SD = 0.68) and human-robot interaction (M = 2.00, SD = 0.95) on a five-point Likert scale, where 5 indicates high familiarity. Their self-reported technical affinity ranged from two to five (M = 3.29, SD = 0.72) on the same scale.

Specific ethics approval was not required by our institution. However, the study complied with the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants at the start of the study, and all collected data were anonymized to ensure confidentiality.

4.3 Independent Variables

The study employs a within-subject design with the independent variable being the force display *technique*. Each *technique* is composed of the factors of visualization *type*, visualization *position*, and *vibrotactile* feedback. The different factors include:

- *Type*: Bar and Arrow force visualization concepts
- *Position*: Placement at Screen, Probe and FOV
- *Vibrotactile*: Presence or absence of vibrotactile feedback

Additionally, the study investigated two baseline conditions. In one condition, the visualization *type* and *position* as well as the *vibrotactile* factors were set to "none," representing no sensory substitution (*technique*: none). In the other condition, only *vibrotactile* feedback was active (*technique*: tactile). The combination of these factor levels resulted in 14 prototypes (2 *types* $\times$ 3 *positions* $\times$ 2 *vibrotactile* + 2 baseline).

4.4 Dependent Variables

As dependent variables, both objective and subjective variables were measured. The first dependent variable was the *task completion time* (TCT), which was measured from the moment participants first reached F_{target} until the completion of the task. Task completion involved visualizing the three geometries, with the study conductor confirming adequate visibility through visual inspection. The other two objective variables concerned the applied force, which was continuously recorded and paused when no interaction was active to prevent constant force exertion during static robot positions from influencing results. To maintain a consistent sampling rate across trials and participants, the force data was interpolated to an equidistant sampling at a sampling rate of 1 kHz. The two dependent variables were the *average absolute deviation* (AAD), calculated as the average absolute deviation from F_{target} at each time point, and the *average maximal applied force* (AMax), computed by averaging the force of local maxima. Local maxima were filtered to include only those with a prominence of 1 N to mitigate the influence of jitter on results. Additionally, participants were asked to rate two questions on a 6-point Likert item (with verbal anchors ranging from "strongly disagree" to "strongly agree"). The questions regarded the *perceived difficulty* ("The currently applied force was difficult to estimate.") and the *perceived continuity* ("I was continuously aware of the magnitude of the applied force.").

4.5 Procedure

The study began with the participants completing a consent form and a questionnaire on demographic data on arrival. They then received a brief introduction to the study objective and the system. The training

session involved participants practicing freehand ultrasound imaging on the phantom, familiarizing themselves with the positioning of the target structures, and practicing the interaction technique. As soon as the participants felt familiar with the imaging procedure and the interaction with the robot, they began one trial for each sensory substitution technique, resulting in a total of 14 trials. To reduce potential ordering effects on study results, the order of the *techniques* was counterbalanced across participants using a 14×14 Latin square. For each trial, participants had the opportunity to practice under the current conditions until they felt confident to proceed. They then began the experimental task, in which they had to successively guide the robot to each geometric structure once. After each task completion, the participants gave their subjective ratings on perceived difficulty and continuity. After completing all trials, a semi-structured interview was conducted. The experiment took an average of 60 minutes.

4.6 Statistical Analyses

All statistical analyses were conducted using R (version 4.1.1). To evaluate effects of the three investigated factors on the observed measures and their respective interaction, the data of the objective variables TCT, AAD, and AMax was first checked for normality using Shapiro-Wilk tests. Three-way repeated measures analyses of variance (ANOVA) were conducted in case this assumption was met. Otherwise, robust three-way ANOVAs for within-subject designs based on trimmed means were calculated (also see [49]). For ordinal data of the Likert-item variables perceived difficulty and perceived continuity, Wilcoxon signed-rank tests for paired samples were conducted to evaluate main effects of factors with two factor levels, i.e., *type* and *vibrotactile*, and Friedman tests were calculated for main effects with more than two factor levels, i.e., *position*. The existence of potential interaction effects was evaluated using robust three-way ANOVAs.

These investigations did not consider data from the two baseline conditions, *none* and *tactile*. To assess them, data from the remaining conditions was aggregated into two points per participant: *visual*, representing the mean for all visual-only conditions, and *multimodal*, representing the mean for all visual and vibrotactile conditions. The resulting data formed a one-factor repeated measures design with four levels. This *modality* factor was then evaluated as described earlier. Normality was checked, followed by repeated measures ANOVAs or robust ANOVAs if violated. Subjective items were analyzed with Friedman tests.

Furthermore, post-hoc pairwise comparisons were conducted for all significant main effects involving more than two factor levels. Pairwise paired t-tests with Bonferroni corrections were conducted for normally distributed objective variables. Robust Yuen's trimmed means tests with p-value adjustments using Hochberg's method were used for non-normal objective data. Pairwise Durbin-Conover tests with Bonferroni corrections were calculated for the subjective measures.

5 RESULTS

5.1 Statistical Results

The following outlines the results of the conducted statistical tests. Emphasis is placed on significant findings. More details on the descriptive and results of the statistical test are provided in the supplementary material. Regarding the tests evaluating the three-factorial design, TCT, and AAD measures were not normally distributed. Hence, robust ANOVAs were conducted. The tests revealed a significant *vibrotactile* main effect on TCT (test statistic $Q = 8.869$, p-value $p = 0.003$, effect size $\delta_t = -0.28$) and a significant *type* main effect on AAD ($Q = 8.068$, $p = 0.005$, $\delta_t = -0.35$). All other main effects showed no statistical significance. Also, neither significant two-way nor significant three-way interaction effects were found. The AMax variable was normally distributed. The repeated measures ANOVA revealed a significant three-way interaction effect between all three factors (test statistic $F = 3.902$, p-value $p = 0.028$, effect size $\eta^2 = 0.03$). No significant main and two-way interaction effects were found on this variable. The subjective measures were not affected by interaction effects. Friedman tests showed significant *position* main effects on subjective difficulty

(test statistic $\chi^2 = 6.237$, p-value $p = 0.044$, effect size $W = 0.15$) and on subjective continuity ($\chi^2 = 7.629$, $p = 0.022$, $W = 0.18$).

Regarding the analyses of the *modality* factor, the TCT data was not normally distributed. Hence, a robust repeated measures ANOVA was conducted. This test revealed a significant effect ($F = 4.804$, $p = 0.016$, $\delta_t = -0.09$). The respective AAD and AMax data were normally distributed. A repeated measures ANOVA showed a significant effect of the *modality* factor on AAD ($F = 7.362$, $p < 0.001$, $\eta^2 = 0.20$). Friedman tests were conducted on subjective difficulty and subjective continuity. These tests also significant effects of the *modality* factor (perceived difficulty: $\chi^2 = 30.429$, $p < 0.001$, $W = 0.48$; perceived continuity: $\chi^2 = 40.215$, $p < 0.001$, $W = 0.64$).

5.2 Qualitative Feedback

For the qualitative feedback analysis, the statements made by the participants during the semi-structured interview were paraphrased and clustered. Clustering focused on statements consistent between at least two participants, with duplicates within participants removed.

In total, 251 individual statements were recorded based on participants' verbal feedback. Of these, 206 were expressed by at least two participants. These statements were condensed into 43 summarizing statements, presented in Table 1.

5.3 Interpretation of Results

5.3.1 Sensory Substitution Modality

The significant results from pairwise comparisons of objective measures across the feedback modalities are depicted in Figure 4. The TCT results show significant differences across feedback modalities, with visual feedback leading to higher TCT than the baseline, indicating that visual feedback slows down interaction. This could be attributed to the additional cognitive load induced by processing visual information, particularly in conjunction with analyzing sonographic images for navigation. Furthermore, TCT was significantly higher with multimodal feedback than without or with visual feedback alone, as indicated by both a significant effect in the analysis of the *modality* factor and a significant main effect of the *vibrotactile* factor in the three-factorial design. Multimodal feedback, specifically the combination of visual and vibrotactile feedback resulting in even slower performance may be attributed to information overload and the heightened concentration required to process the combined feedback, as reported by participants during the semi-structured interviews. In terms of AAD, the results indicate a significantly higher force deviation for tactile feedback alone compared to multimodal and visual feedback. This discrepancy may originate from the slower perception of information during tactile feedback, particularly evident when manipulating the probe, where force changes take longer to be perceived relative to the movement speed. Moreover, participants reported difficulties in recognizing the pattern of tactile feedback, potentially leading to confusion between patterns representing forces too high or too low. This confusion might have resulted in participants inadvertently adjusting the force in the wrong direction. Notably, there was no significant difference in AAD between the multimodal and visual feedback conditions, suggesting a tendency among participants to disregard vibrotactile feedback when visual feedback was present. This is consistent with feedback during the interview.

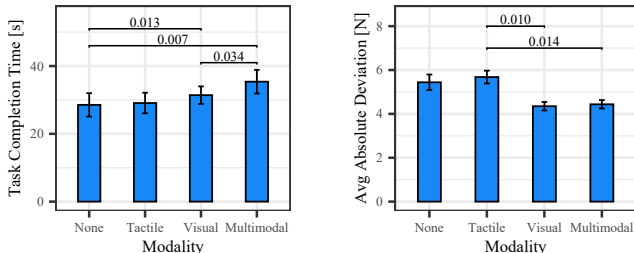


Fig. 4: Objective measures across feedback modalities

The subjective measures reveal improvements in perceived difficulty and continuity for all feedback conditions compared to the baseline no-feedback condition (see Figure 5). Moreover, results indicate that both visual and multimodal feedback significantly reduce perceived difficulty and enhance perceived continuity compared to vibrotactile feedback alone. These findings align with the results for AAD and suggest that ambiguities regarding the vibration pattern may contribute to lower user satisfaction with vibrotactile feedback. In contrast, visual feedback might provide a more direct and clear information representation, leading to enhanced performance and higher user satisfaction. Notably, while multimodal and visual feedback significantly increase completion time, they reduce perceived difficulty. This apparent discrepancy may be explained by providing feedback reducing perceived difficulty by increasing participants' ability to assess the force applied. However, this may also slow down the interaction as participants need to respond to the feedback to adjust their force application.

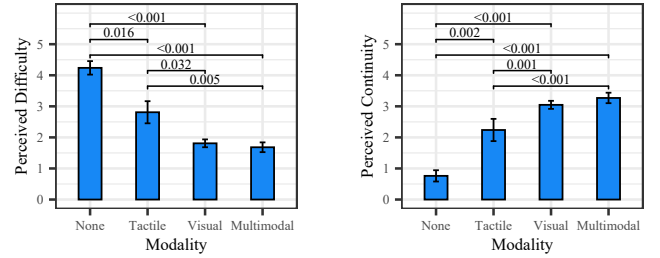


Fig. 5: Subjective measures across feedback modalities.

5.3.2 Visualization Type

Significant main effects regarding visualization type are depicted in Figure 6. The pairwise comparison indicates that the bar visualization resulted in significantly lower AAD than the arrow. This could be due to the discrete coloring and subdivision, which effectively highlights F_{target} and is consistent with user feedback describing the bar as precise and sensitive. Despite participants' reported preference for the arrow's gradual changes in size and color, this feature may have contributed to a less accurate perception of small force deviations. Especially as the arrow becomes increasingly smaller as the currently applied force approaches F_{target} , making subtle changes more difficult to identify.

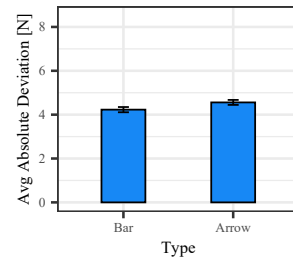


Fig. 6: Main effects of *Type* factor.

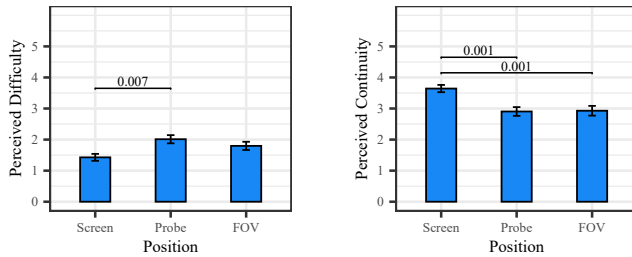
5.3.3 Visualization Position

For the visualization position, significant main effects were found for perceived difficulty and perceived continuity (see Figure 7). The pairwise comparison revealed the screen position having a significantly lower perceived difficulty in maintaining F_{target} and a significantly more continuous feedback perception compared to the probe position. This could be explained by the statements given during the semi-structured interview where, although some participants reported looking at the probe to observe the robot's movements, the majority stated mainly looking at the sonographic image on the screen and therefore preferring the feedback on the screen. Regarding FOV positioning, pairwise comparisons showed significantly less continuous feedback perception

Table 1: Summary and frequency of statements (#) received during the semi-structured interview

Type	Position	Vibrotactile
Bar: + easy to read (2) + correct force precisely recognizable (2) + intuitive proportions (2) - too sensitive due to small green area (4) - high sensitivity causes changes to be overestimated and corrected too strongly (4)	Screen: + helpful when observing US image (17) - outside of view when looking at probe (2)	+ intuitive pattern (7), helpful(6), correct force easily recognizable (2), pleasant (2) + draws attention to changes (6) + helpful when visualization out of view (4) + supports visual feedback (4) + extra sensory channel alongside visual input (4) - pattern ambiguous (9) and needs concentration (6) - requires training to understand (6) - more focus on visualization, if present (6) - ignored with habituation (3) - leads to information overload (3) - not helpful (2), disturbing (2), and stressful (2) - information delivery too slow (2) - difficult as sole feedback modality (2) • preferable only as alarm for excessive force (6) • increasing frequency with force better pattern (2)
Arrow: + intuitive (11), continuous (6), precise (2) + visualization of direction helpful (9) + peripheral changes perceptible (8) + continuous color scale striking (2) - force direction confusing (3)	Probe: + helpful when looking at probe to estimate robot movement (6) - out of view when looking at screen (11)	
	FOV: + helpful for viewing both screen and probe (3) - requires looking at additional position (6) - is not in focus (6) - position in world not fixed confusing (3) - disturbing (2)	

compared to screen placement. Despite the visualization being theoretically visible from any direction due to its body-stabilized position, participants found it out of focus, requiring a conscious gaze shift. They also reported confusion with the dynamic positioning, which likely further reduced the perceived continuity of force perception.

Fig. 7: Main effects of *Position* factor.

5.3.4 Interactions between Factors

In addition, a three-way interaction was observed for AMax (see Figure 8). The effect suggests that without vibrotactile feedback, particularly the bar visualization on the screen performs worse than the other combinations of *type* and *position*. This finding seems to be in contrast to the subjective feedback, where a preference for screen position was described. However, a possible explanation could be that although the screen display enables continuous perception, it primarily engages peripheral vision. Since the bar visualization employs a discrete color scheme, changes leading to maxima may not be immediately perceived if the focus is not directly on the visualization, potentially resulting in higher maxima. Other visualization placements may require a more deliberate shift in focus, leading to a more accurate perception of changes.

When vibrotactile feedback is available, this trend shifts, with the bar visualization at the screen position performing better than at the probe position. This aligns with the results for perceived difficulty, where recognizing the applied force at the probe position is rated as more challenging. Vibrotactile feedback, therefore, appears to enhance the perception of the bar visualization on the screen by potentially alerting participants to changes and guiding their attention to deliberately shift toward the visualization. The influence of vibrotactile feedback and visualization position on AMax for arrow visualization appears to be less pronounced. This may be attributed to changes in size and color being easily perceived even when only peripherally visible, making AMax relatively independent of other factors for arrow visualization. This observation is consistent with participants' feedback, which described the arrow's changes as perceptible peripherally and noted the striking color scale.

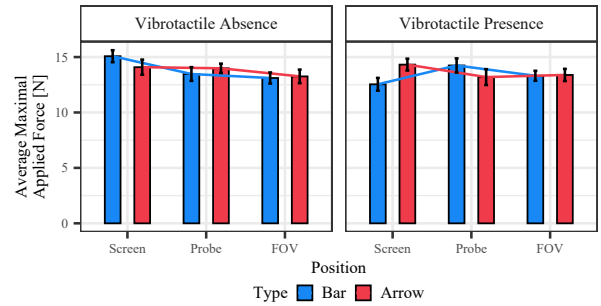


Fig. 8: Three-way interaction effect for AMax.

6 DISCUSSION

The study results indicate that while participants found vibrotactile feedback intuitive and helpful, force error, perceived difficulty, and continuity were primarily dependent on the presence of visual feedback. This indicates a tendency for participants to prioritize visual over vibrotactile sensory substitution, which aligns with the qualitative feedback indicating neglect of vibrotactile cues. These results are consistent with previous research by Howard et al. [19], reporting superior performance with visual compared to vibrotactile feedback. Additionally, in line with Jonetzko et al. [21], we observed a reduced efficiency with multimodal feedback, indicating information overload. A major limitation of our study is employing only one type of vibrotactile feedback. The scaling approach was selected based on literature highlighting its suitability for vibrotactile feedback and its assumed comparability to the information density of visual feedback. However, discrepancies between visual and vibrotactile scaling may have influenced user perception. Additionally, previous work suggests that while non-linear scaling can improve accuracy, it may increase perceived difficulty and completion time [17]. Therefore, further investigation is needed into the scaling and the alignment of different scaling methods and their impact on task performance. Unlike our relative force magnitude vibrotactile pattern, Chan et al. [6] employed constant patterns to indicate force direction alone, achieving superior performance compared to visual feedback. Given that participants expressed a preference for constant vibration for excessive force feedback, or increasing frequency with increasing force, exploring these alternatives could yield different results.

Regarding visualization type, study results suggest that bar visualization facilitates more accurate force application, whereas participants reported better peripheral perceptibility for arrow visualization. These differences may be attributed to different design elements, such as the challenge of deriving force deviation from small arrow visualizations close to F_{target} , or the advantage of continuous color scale. However, us-

ing two distinct approaches with multiple variables - such as 2D vs 3D and discrete vs continuous color and shape changes - makes it difficult to isolate the impact of each variable on the results. Therefore, future research could focus specifically on these aspects to gain deeper insights. The visualizations additionally differed in the information displayed, with the arrow indicating the force direction. Although the participants described the directional display as helpful, it didn't significantly impact quantitative measurements, likely due to task simplicity. Future studies with more complexity may yield different outcomes. While the study results indicate that the bar visualization allowed for smaller deviations from F_{target} , some participants noted that the threshold for the green area was too small, making the bar feel overly sensitive. Although the threshold was designed to enhance precise perception of changes around the target force, the rapid changes in indicator color could lead to difficulties in force interpretation. Therefore, future work could evaluate the impact of increasing the size of the visualization threshold on both the accuracy of force application and user feedback, to determine whether user satisfaction can be improved without compromising or even enhancing accuracy. Future work could additionally explore and compare scaling visual approaches, such as logarithmic and linear scaling as done in Heinrich et al. [17], to assess their impact on task performance.

Results indicate the suitability of visualization placement is dependent on the operator's focus during the task. While some participants reported looking at the probe during interaction to perceive the robot's reaction, the majority indicated looking at the probe for initial orientation but then focusing primarily on the screen. The main factor contributing to the advantage of screen placement is therefore the distance between the sonographic image and the probe. However, an alternative approach could involve displaying the ultrasound image in AR near the probe [35], potentially integrating the probe and screen visualizations presented here.

While the sensory substitution technique design is based on literature findings, qualitative feedback suggests possible design flaws that may have influenced study results. However, since the main feedback primarily concerns the nature of the techniques rather than specific design decisions themselves, we believe that these flaws do not diminish the overall significance of our findings. Some of these findings, such as the risk of tactile clutter reducing comprehension and sensory overload with vibrotactile feedback, have been previously investigated in studies focusing on general multimodal feedback design [47]. Nevertheless, our study provides valuable insights by offering empirical evidence, replication, and validation for a new context: force perception in human-robot interaction.

The study used an optical see-through HMD for feedback visualization, which has a high implementation cost. This was chosen for its ability to ensure consistent visual properties across different visualization placements, allowing comparability. However, based on our results showing the benefits of placing the visual feedback adjacent to the ultrasound image, this feedback could be implemented more cost-effectively by integrating it into the display. While the HMD in our study also supported hand tracking and interaction, alternative feedback visualization approaches may be more viable in scenarios where different input methods are used for robot control.

To compare unimodal and multimodal feedback, two baseline conditions were included. However, more data was available for the other conditions than the baseline, leading to an unbalanced design. To ensure a balanced sample for the statistical tests, the data was aggregated. Since no interaction effects were found for TCT, AAD, perceived difficulty, and continuity, we believe that this aggregation did not result in information loss, and the results remain valid. Notably, only 21 participants were included in the study. While the repeated measures per factor level increase the power of our analysis due to the factorial design, the interaction effects have lower power, and additional participants would be required to robustly examine these effects.

6.1 Limitations

While the results provide valuable insights into sensory substitution's impact on force perception in robotic ultrasound, several limitations

should be acknowledged. Firstly, the study utilized an abstract phantom, designed to mimic the acoustic properties of the human body but lacking the complexities of clinical sonographic tasks. The simplicity of the task may have impacted participants' mental capacity to focus on force display. Additionally, only one phantom with uniform stiffness was used in this study. Applying the system to softer tissues may require manual adjustment of force values or deriving local tissue stiffness (e.g., through palpation) to dynamically adjust the force parameters. Future research should aim to validate our findings in clinical contexts with more realistic tasks. Furthermore, we recruited participants using random sampling, primarily focusing on medical background, without considering gender, age, or ethnicity, which may affect the generalizability of our findings. While the sample reflects the demographic distribution at the local university, future studies should aim to include more diverse populations to enhance the validity. Additionally, this study employed purely manual robot control. Future research could explore hybrid force-position control to compare manual and autonomous systems and assess if sensory substitution improves transparency and confidence in autonomous systems.

The final limitation is the specificity of our study, which focused solely on robotic ultrasound applications, thus limiting the generalizability to other human-robot interaction contexts. This, as well as the focus on visualizations that allow flexibility in placement led to the exclusion of other potentially promising approaches. While our results indicate that visualizations placed at the site of force application, namely the probe, were associated with higher perceived difficulty and less continuous force perception, this placement could be optimized with more complex approaches, such as surface deformation effects [39] or offsetting a virtual proxy to represent force [44]. Future work could compare these visualizations with the effective techniques identified in this study to gain deeper insights.

While our study focused on robotic ultrasound, some findings may also apply to other contexts where direct force perception is limited, such as laparoscopic procedures or industrial robotic tasks like polishing, or robotic grasping. While some adjustments to our presented methods might be necessary for these different contexts, the core principles are transferable. The force values used during our study are relatively small, which allows for changes to be easily perceptible on a linear scale. However, for tasks requiring larger forces, exploring different scaling methods may be necessary to accurately represent fluctuations near the target force. While the force visualization in our study aimed to support the reaching of a certain target force, similar to robotic surgery tasks such as tightening knots [2] or industrial tasks such as polishing [9], there are scenarios such as grasping [21] where absolute force values hold greater significance. In such cases, the methods employed in our study can be easily adjusted by setting $F_{\text{target}} = 0$, allowing feedback to represent absolute force values. Regarding the visualization placements, the focus on a monitor during co-located interaction is specific to robotic ultrasound. However, while placement at the probe and on the screen is application-specific, general insights can be gained from world- and head-stabilized visual sensory substitutions. Although we therefore believe that the general approaches of our work could apply to other use cases, future research should aim to test the suitability of these concepts for different tasks.

6.2 Implications

In summary, we propose the following design implications, based on our study results, for integrating sensory substitution to enhance force perception during human-robot interaction, which can be generalized beyond robotic ultrasound:

Employ visual sensory substitution to enhance force perception: Our findings suggest that any form of sensory substitution significantly reduces perceived task difficulty compared to no feedback, with the presence of visual feedback being particularly effective in lowering force errors and difficulty while enabling continuous perception of applied force. Regarding multimodality, care should be taken in feedback design, as the combination of multiple modalities can reduce task efficiency by increasing task duration. However, incorporating discrete vibrotactile feedback to complement visual feedback by providing

alerts for changes may be beneficial in optimizing performance.

Utilize bar visualizations to accurately depict force: Our study results indicate that representing force magnitude with a bar visualization, where the size is proportional to force magnitude and the deviation from a target force is indicated by discrete coloring, offers a clear representation and helps minimize force errors. However, if the visualization needs to be peripherally perceivable, mapping force error onto size and incorporating gradual color changes could enhance perceptibility.

Employ world-stabilized placement of virtual content near task-relevant locations: Our findings suggest that placing virtual content on the screen displaying ultrasound data reduces the perceived difficulty and enhances continuous perceptibility. Qualitative feedback indicates user preference for world-stabilized visualization placement since the dynamically changing position of the head-stabilized visualization in the environment as well as its positioning not being directly in the line of sight require a conscious shift of attention, complicating perceptibility.

7 CONCLUSION

In this study, we investigated different AR concepts for sensory substitution to enhance force perception during robotic ultrasound. These concepts differed in modality, visualization design, and placement. We evaluated the suitability of unimodal and multimodal visual and vibrotactile approaches through a user study assessing task completion time, average maximal applied force, average force error, and subjective feedback. Our results indicate that any form of feedback improves force perception, with visual feedback being particularly effective in reducing force errors and being preferred by users. Specifically, a 2D bar visualization, changing height and color, was found to be better at reducing force errors compared to a 3D arrow visualization displaying relative force through gradual color, direction, and size changes. Moreover, users subjectively preferred the display of feedback on the screen displaying the sonographic image over positioning it in their field of view or at the ultrasound probe. These results guided the formulation of guidelines for sensory substitution in human-robot interaction and are an important step toward force perception in robotic ultrasound. Future work should focus on assessing the transferability of our results to other use cases in human-robot interaction.

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